Homework #1 (Submit before 16-Mar)

❖ Ch1_No.2

If the continuous output power of lasers is 1 W when λ=10 μm, λ=500 nm, or v=3000 MHz, respectively, what is the number of particles that transit from the upper energy level to the bottom energy level per second?

❖ Ch1_No.3

Suppose the energy level of the laser device is E_2 and E_1 where f_2=f_1, the corresponding frequency is v (wavelength is λ), and the particles number at each energy level is n_2 and n_1, respectively.

a) When v=3000 MHz, T= 300 K, then n_2/n_1 =?

b) When λ=1 μm, T= 300 K, then n_2/n_1 =?

c) When λ=1 μm, n_2/n_1 =0.1, then T=?

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❖ Ch1_No.6

The transition probability of the spontaneous emission from the energy level of a molecule E_4 to the three lower energy levels E_1, E_2, and E_3 is A_{43}=5*10^7 s^{-1}, A_{42}=1*10^7 s^{-1}, and A_{41}=3*10^7 s^{-1} respectively. Try to find the spontaneous emission lifetime τ_{s4} of E_4.

If the lifetime of each energy level is τ_1=5*10^{-7} s, τ_2=6*10^{-9} s, τ_3=1*10^{-8} s, and τ_4= τ_{s4}. when continuously exciting E_4 and reaching the steady-state, try to find the ratio of the number of particles between the corresponding energy levels n_1/n_4, n_2/n_4, and n_3/n_4, and then point out which of the two energy levels has achieved the population inversion.